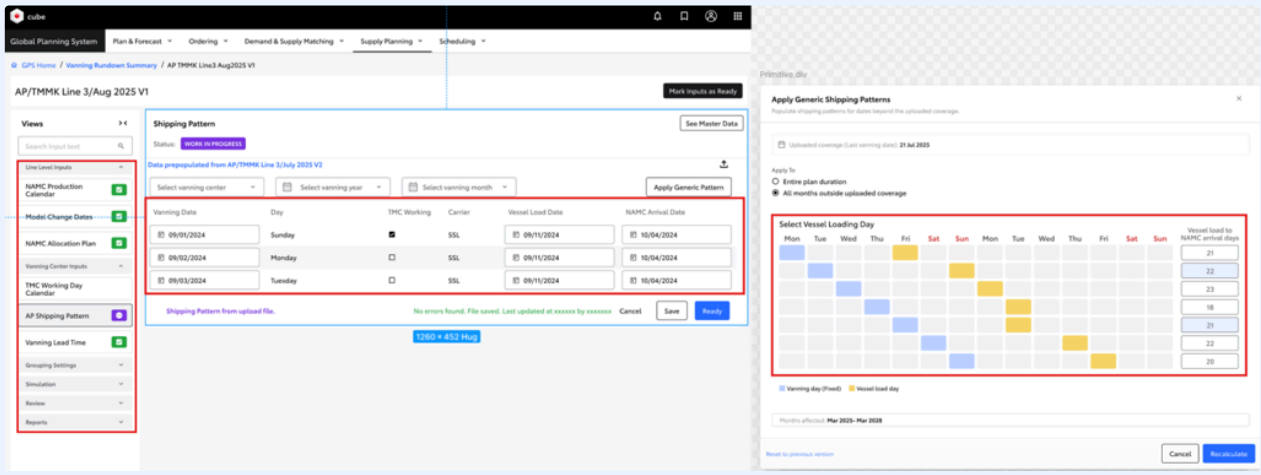
SPPL-195: Shipping Pattern Get API technical workflow

Jira Ticket: SPPL-195: BE | Create technical workflow for Shipping Pattern Get & Post APIs on parameter inputs page including DB details ACCEPTED

This page provides comprehensive information about the GET API for Shipping Pattern on the Input params page, to view shipping details.



Supply Planning GET Shipping Pattern API Details:

1. **Hosted URL**: https://api.spln.cube<env>.toyota.com/<env>
2. **API Name:** <HOSTED_URL>/v/1/1/sp/shipping-pattern
3. **Path Param**: N/A
4. **Query Param**: ? scenarioId=uniqueScenarioId&vanningCenter=NH&month=Jan&year=2025&userEmail=userEmail
5. **Method:** GET
6. **Authorization**: Bearer token (**required**)
7. **Lambda Name**: spln-<env>-view-shipping-pattern-v1-1
8. **API Request Body**: N/A
9. **API Response:**
   - **Success response with status 200:**

```
{
  "scenarioId": "uniqueScenarioId",
  "vanningCenter": "NH",
  "month": "Jan",
  "year": "2025",
  "data": [
    {
      "vanningDate": "2025-01-01",
```

```
      "dayOfWeek": "Monday",
      "isWorkingDay": true,
      "carrier": "SSL",
      "loadDate": "2025-01-11",
      "arrivalDate": "2025-02-10"
    },
    {
      "vanningDate": "2025-01-02",
      "dayOfWeek": "Tuesday",
      "isWorkingDay": true,
      "carrier": "SSL",
      "loadDate": "2025-01-11",
      "arrivalDate": "2025-02-10"
    }
  ],
  "genericPattern": {
    "entirePlanDuration": true,
    "data": [
      {
        "vanningDay": 0,
        "loadDay": 4,
        "loadToArrivalDays": 14
      },
      {
        "vanningDay": 1,
        "loadDay": 7,
        "loadToArrivalDays": 20
      },
      {
        "vanningDay": 10,
        "loadDay": 13,
        "loadToArrivalDays": 24
      }
    ]
  },
  "scenarioSteps": {
    "Line Level Inputs": {
      "Model Change Dates": "Completed",
      "NAMC Allocation Plan": "Completed",
      "NAMC Production Calendar": "Completed"
    },
    "Vanning Center Inputs": {
      "Shipping Pattern": "Completed",
      "Vanning Lead Time": "Completed",
      "TMC Working Day Calendar": "Completed"
    },
    "Grouping Settings": {
      "Grouping": "Completed",
      "Min Max Stock": "In Progress",
      "Fluctuation Allowance": "Not Started"
    },
    "other": [
      "Simulation",
      "Review",
      "Reports"
    ]
  }
}
```


- **Error Response:**
  - **any internal server error occurred with status 500:**


```
{
  "errorMessage": "Internal Server Error"
}
```

- **any unauthorized error occurred with status 401 (Managed by the AWS API Gateway integrated Custom Authorizer):**

```
{
  "message": "Unauthorized"
}
```

- **If request is in-valid with status 400:**

```
{
  "errorMessage": [<"ValidationError: validation error message”>]
}
```

- **List of possible validation error message:**
  1. "ValidationError: scenarioId is required and must be a uuid."
  2. “ValidationError: vanningCenter is required and must be a string.”
  3. “ValidationError: month is required and must be a 3-character long string.”
  4. “ValidationError: year is required and must be a 4-character long string.”
  5. "ValidationError: Scenario doesn't exist."
  6. “ValidationError: userEmail is required and must be a string.”
  7. “ValidationError: Invalid userEmail.”

10. **Table:**
    a. **scenarios**
    b. **shipping_pattern**
    c. **generic_pattern**
    d. **scenario_step_status**

SP Shipping Pattern Get API Workflow:

- **Technical workflow:**
  - Description:
    i. The Supply Planning user logs in to the application.
    ii. After the user provides creates a scenario, opens it and clicks on Shipping Pattern, a request is sent to the SP Get Shipping Pattern endpoint: ***sp/shipping-pattern*** with parameters: scenarioId=uniqueScenarioId&vanningCenter=NH&month=Jan&year=2025&userEmail=userEmail
    iii. After successful authentication, the **spln-view-*shipping-pattern*-v1-1** lambda is triggered by the SP view scenario Shipping Pattern endpoint: ***sp/shipping-pattern***.

If authentication fails, an error with status 401 (*refer: error response-**any unauthorized error occurred with status 401***) will be returned and sent to client.

iv. After user authentication has been confirmed, the **spln-view-*shipping-pattern*-v1-1** lambda will perform validation on the request payload. If payload is valid will proceed to next step else throw validation error (*refer: error response**: Section if request is in-valid with status 400 and List of possible validation error message***). Below query is used to check if provided scenario is valid:

```
select * from supply_planning.scenarios where scenario_id = :scenarioId and
is_active = true;
```

v. After the request payload is successfully validated, the **spln-view-*shipping-pattern*-v1-1** lambda execute below queries to get scenario Shipping Pattern data from the ***shipping_pattern*** & **generic_pattern** tables. Ensure to sort generic pattern data by vanningDay.

```
select * from supply_planning.shipping_pattern where scenario_id = :scenarioId
and vanning_center = :vanningCenter
and vanning_date like '<year-month>%' and tmc_working = true;
select * from supply_planning.generic_pattern where scenario_id = :scenarioId
and vanning_center = :vanningCenter;
```

vi. Below query is used to fetch view scenario details (if status data doesn't exist for a scenario, then insert the details): ref(SPPL-162: View Scenario Menu technical workflow document)

```
select * from supply_planning.scenario_step_status where scenario_id =
:scenarioId;
```

vii. The retrieved data is formatted as shown below before returning the response to the client with a status code of 200:

1. If the Shipping Pattern data for scenario is available:

```
{
  "scenarioId": "uniqueScenarioId",
  "vanningCenter": "NH",
  "month": "Jan",
  "year": "2025",
  "data": [
    {
      "vanningDate": "2025-01-01",
      "dayOfWeek": "Monday",
      "isWorkingDay": true,
      "carrier": "SSL",
      "loadDate": "2025-01-11",
      "arrivalDate": "2025-02-10"
    },
    {
      "vanningDate": "2025-01-02",
      "dayOfWeek": "Tuesday",
      "isWorkingDay": true,
      "carrier": "SSL",
      "loadDate": "2025-01-11",
```

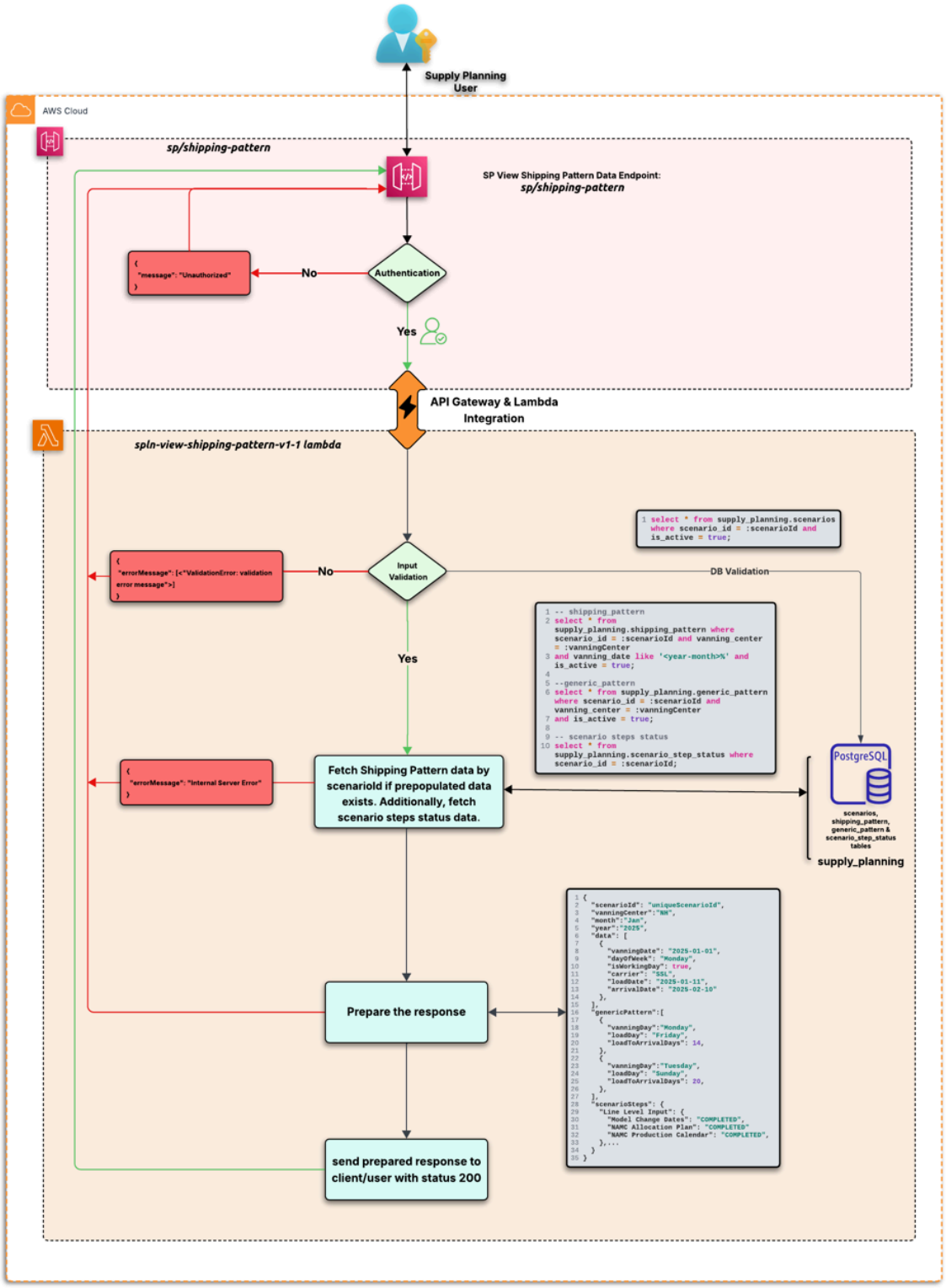
```
      "arrivalDate": "2025-02-10"
    }
  ],
  "genericPattern": {
    "entirePlanDuration": true,
    "data": [
      {
        "vanningDay": 0,
        "loadDay": 4,
        "loadToArrivalDays": 14
      },
      {
        "vanningDay": 1,
        "loadDay": 7,
        "loadToArrivalDays": 20
      },
      {
        "vanningDay": 10,
        "loadDay": 13,
        "loadToArrivalDays": 24
      }
    ]
  },
  "scenarioSteps": {
    "Line Level Inputs": {
      "Model Change Dates": "Completed",
      "NAMC Allocation Plan": "Completed",
      "NAMC Production Calendar": "Completed"
    },
    "Vanning Center Inputs": {
      "Shipping Pattern": "Completed",
      "Vanning Lead Time": "Completed",
      "TMC Working Day Calendar": "Completed"
    },
    "Grouping Settings": {
      "Grouping": "Completed",
      "Min Max Stock": "In Progress",
      "Fluctuation Allowance": "Not Started"
    },
    "other": [
      "Simulation",
      "Review",
      "Reports"
    ]
  }
}
```

2. If Shipping Pattern data is not available:

```
{
  "scenarioId": "uniqueScenarioId",
  "vanningCenter":"NH",
  "month":"Jan",
  "year":"2025",
  "data": [],
  "genericPattern":[],
  "scenarioSteps": {
    "Line Level Inputs": {
      "Model Change Dates": "Completed",
      "NAMC Allocation Plan": "Completed",
      "NAMC Production Calendar": "Completed"
    },
    "Vanning Center Inputs": {
      "Shipping Pattern": "Completed",
      "Vanning Lead Time": "Completed",
      "TMC Working Day Calendar": "Completed"
```

```
    },
    "Grouping Settings": {
      "Grouping": "Completed",
      "Min Max Stock": "In Progress",
      "Fluctuation Allowance": "Not Started"
    },
    "other": [
      "Simulation",
      "Review",
      "Reports"
    ]
  }
}
```

viii. If there is an error while fetching the data, an “Internal Server Error” with a status code of 500(*refer: error response-**any internal server error occurred with status 500***) will be returned. Otherwise, a successful response with a status code of 200 will be sent.



SP Shipping Pattern Get API Techncial workflow

Note:

We will be utilizing this API for both to fetch file data and previous rundown data .

Queries: